Assignment 3: Data Exploration

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Exploration.

Directions

1. Rename this file `<FirstLast>_A03_DataExploration.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change "Student Name" on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Assign a useful **name to each code chunk** and include ample **comments** with your code.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.
7. After Knitting, submit the completed exercise (PDF file) to the dropbox in Canvas.

TIP: If your code extends past the page when knit, tidy your code by manually inserting line breaks.

TIP: If your code fails to knit, check that no `install.packages()` or `View()` commands exist in your code.

Set up your R session

1. Load necessary packages (tidyverse, lubridate, here), check your current working directory and upload two datasets: the ECOTOX neonicotinoid dataset (ECOTOX_Neonicotinoids_Insects_raw.csv) and the Niwot Ridge NEON dataset for litter and woody debris (NEON_NIWO_Litter_massdata_2018-08_raw.csv). Name these datasets "Neonics" and "Litter", respectively. Be sure to include the sub-command to read strings in as factors.

```
#R session set-up

#loading packages
library(tidyverse); library(lubridate); library(here)

#checking working directory
getwd()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```r
here()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```r
#reading in datasets
Neonics <- read.csv(
  file = here('Data/Raw/ECOTOX_Neonicotinoids_Insects_raw.csv'),
  stringsAsFactors = TRUE
)

Litter <- read.csv(
  file = here('Data/Raw/NEON_NIWO_Litter_massdata_2018-08_raw.csv'),
  stringsAsFactors = TRUE
)
```

Learn about your system

2. The neonicotinoid dataset was collected from the Environmental Protection Agency's ECOTOX Knowledgebase, a database for ecotoxicology research. Neonicotinoids are a class of insecticides used widely in agriculture. The dataset that has been pulled includes all studies published on insects. Why might we be interested in the ecotoxicology of neonicotinoids on insects? Feel free to do a brief internet search if you feel you need more background information.

   Answer: Neonicotinoids are insecticides developed to control agricultural pests and protect crops. However, their effects are indiscriminate and have been shown to damage populations of native and beneficial insects as well. Learning about the impacts of neonics on a broad array of species can inform ecologically responsible decisionmaking in agriculture.

3. The Niwot Ridge litter and woody debris dataset was collected from the National Ecological Observatory Network, which collectively includes 81 aquatic and terrestrial sites across 20 ecoclimatic domains. 32 of these sites sample forest litter and woody debris, and we will focus on the Niwot Ridge long-term ecological research (LTER) station in Colorado. Why might we be interested in studying litter and woody debris that falls to the ground in forests? Feel free to do a brief internet search if you feel you need more background information.

   Answer: Leaf litter and woody debris can correspond to a number of forest variables including composition, biomass, and productivity; serve as habitat for biodiversity across taxa; and represent a crucial step in the cycling of nutrients among living plant matter, soils, streams, and other organisms. These factors make them information-rich ecosystem descriptors.

4. How is litter and woody debris sampled as part of the NEON network? Read the NEON_Litterfall_UserGuide.pdf document to learn more. List three pieces of salient information about the sampling methods here:

   Answer: 1. Litter and debris are collected using elevated and ground traps, respectively, which are designed in accordance with the methods of the Smithsonian Tropical Research Insitute Center for Tropical Forest Science. 2. Sampling is done on plots that are located within specifically defined airsheds. 3. Samples are taken from ground traps once per year, and from elevated traps at a frequency determined by the type of vegetation present.

Obtain basic summaries of your data (Neonics)

5. What are the dimensions of the dataset?

```
#Neonics dimensions
dim(Neonics)
```

```
## [1] 4623   30
```

6. Using the **summary** function on the "Effect" column, determine the most common effects that are studied. Why might these effects specifically be of interest? [Tip: The `sort()` command is useful for listing the values in order of magnitude…]

```
#Summary of Neonics Effects, sorted
sort(summary(Neonics$Effect))
```

```
##       Hormone(s)        Histology       Physiology          Cell(s)
##                1                5                7                9
##     Biochemistry     Accumulation     Intoxication    Immunological
##               11               12               12               16
##       Morphology           Growth        Enzyme(s)         Genetics
##               22               38               62               82
##        Avoidance      Development     Reproduction Feeding behavior
##              102              136              197              255
##         Behavior        Mortality       Population
##              360             1493             1803
```

Answer: The effects with the largest number of observations are Population (1803) and Mortality (1493). These effects may have the greatest implications for ecological health when using neonics.

7. Using the **summary** function, determine the six most commonly studied species in the dataset (common name). What do these species have in common, and why might they be of interest over other insects? Feel free to do a brief internet search for more information if needed.[TIP: Explore the help on the `summary()` function, in particular the `maxsum` argument…]

```
#Summary of Neonics Species.Common.Name (6 most common)
summary(Neonics$Species.Common.Name,maxsum=7)
```

```
##            Honey Bee        Parasitic Wasp Buff Tailed Bumblebee
##                  667                   285                   183
##   Carniolan Honey Bee            Bumble Bee      Italian Honeybee
##                  152                   140                   113
##              (Other)
##                 3083
```

Answer: The six most commonly observed species are Honey Bee (667), Parasitic Wasp (285), Buff Tailed Bumblebee (183), Carniolan Honey Bee (152), Bumble Bee (140), and Italian Honeybee (113). These are all species considered beneficial to agriculture due to their pollination and predatory behaviors.

8. Concentrations are always a numeric value. What is the class of `Conc.1..Author.` column in the dataset, and why is it not numeric? [Tip: Viewing the dataframe may be helpful…]

```r
#Finding class of Conc.1..Author. column
class(Neonics$Conc.1..Author.)
```

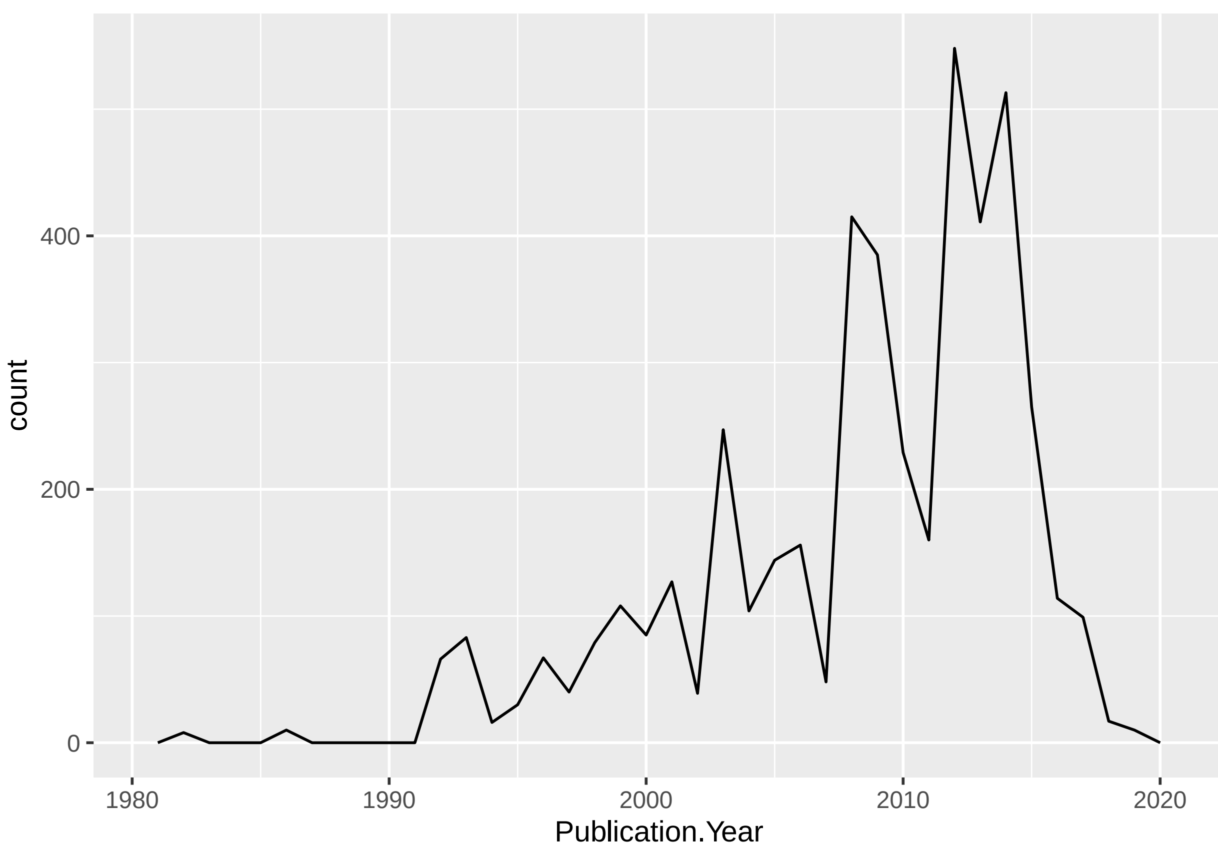
```
## [1] "factor"
```

> Answer: The class of the Conc.1..Author column is factor. This is because some cells in the column contain non-numeric symbols, such as "NR" and "~10." Consequenly, R interprets the entire column as string data, which gets reclassified as factor data upon reading via the "stringsAsFactors" command.

Explore your data graphically (Neonics)

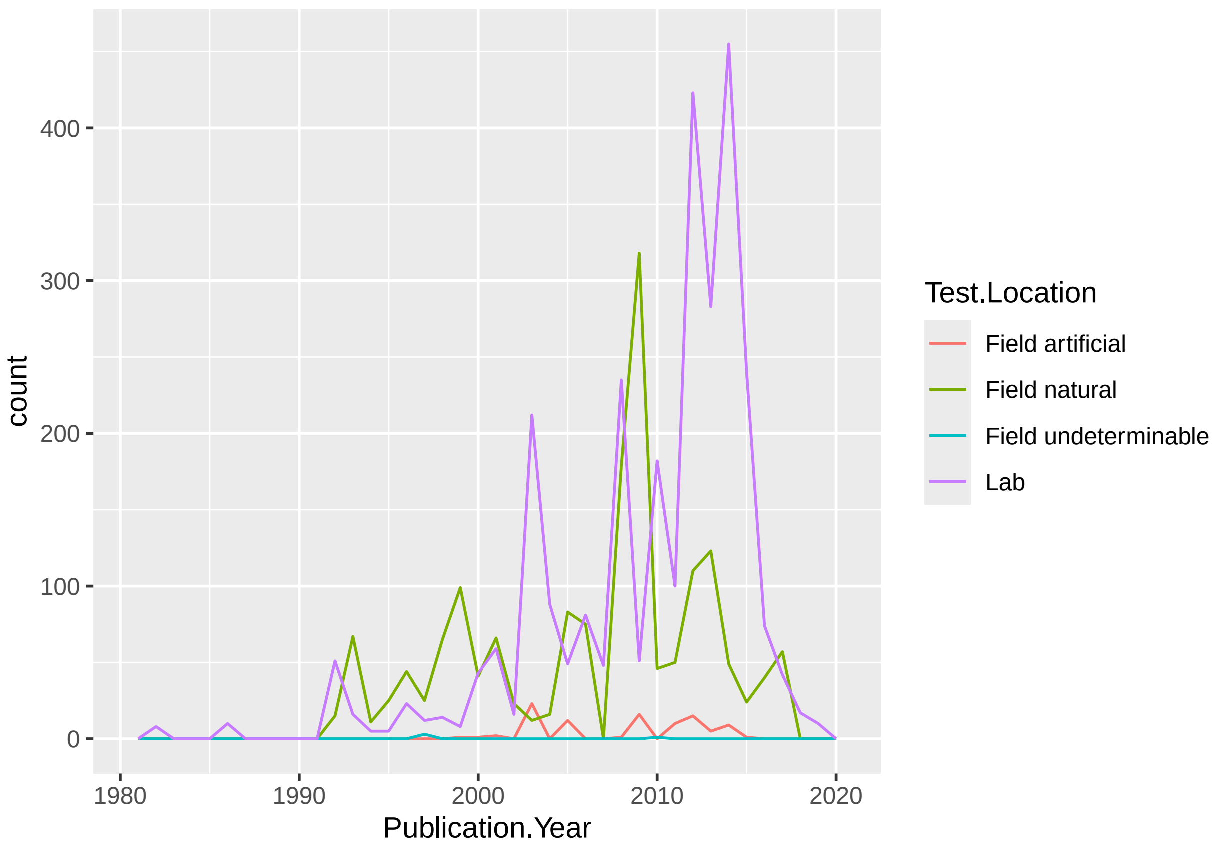
9. Using `geom_freqpoly`, generate a plot of the number of studies conducted by publication year.

```r
#Plotting frequency polygon of count vs. publication year
ggplot(Neonics) +
  geom_freqpoly(aes(x=Publication.Year), binwidth=1)
```



10. Reproduce the same graph but now add a color aesthetic so that different Test.Location are displayed as different colors.

```r
#Replotting count vs. publication year, with test location by color
ggplot(Neonics) +
  geom_freqpoly(aes(x=Publication.Year,color=Test.Location),binwidth=1)
```



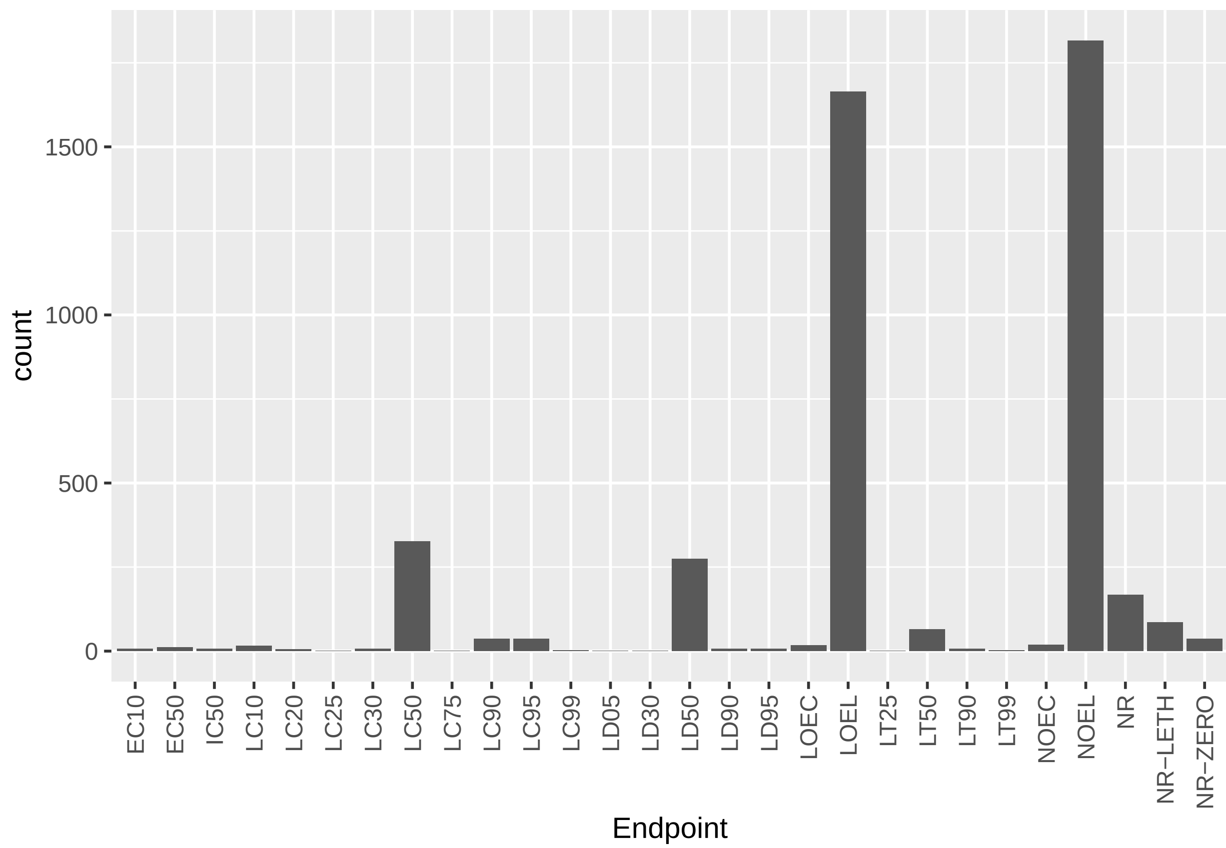
Interpret this graph. What are the most common test locations, and do they differ over time?

> Answer: The most common test location is Lab, followed by Field natural. Overall, Lab and Field natural tests were roughly equally common until about 2011. At this point, Lab tests began surging, reaching a peak of about 700 tests just before 2015. By contrast, Field natural tests declined during the same period.

11. Create a bar graph of Endpoint counts. What are the two most common end points, and how are they defined? Consult the ECOTOX_CodeAppendix for more information.

[**TIP**: Add `theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))` to the end of your plot command to rotate and align the X-axis labels…]

```r
#Plotting endpoint counts as bar graph
ggplot(Neonics) +
  geom_bar(aes(x=Endpoint)) +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))
```



Answer: The most common endpoints are: NOEL (no observable effect level), defined by the highest chemical concentration resulting in no observable difference from control responses; LOEL (lowest observable effect level), defined by the lowest chemical concentration resulting in significantly different responses from controls.

Explore your data (Litter)

12. Determine the class of collectDate. Is it a date? If not, change to a date and confirm the new class of the variable. Using the `unique` function, determine which dates litter was sampled in August 2018.

```
#Assessing dates

#Finding class of collectDate
class(Litter$collectDate)
```

```
## [1] "factor"
```

```
#Reclassifying collectDate as date
Litter$collectDate <- ymd(Litter$collectDate)

#Confirming date class
class(Litter$collectDate)
```

```
## [1] "Date"
```

```r
#Finding unique collection dates
unique(Litter$collectDate)
```

```
## [1] "2018-08-02" "2018-08-30"
```

13. Using the `unique` function, list the different plotIDs sampled at Niwot Ridge. How is the information obtained from `unique` different from that obtained from `summary`?

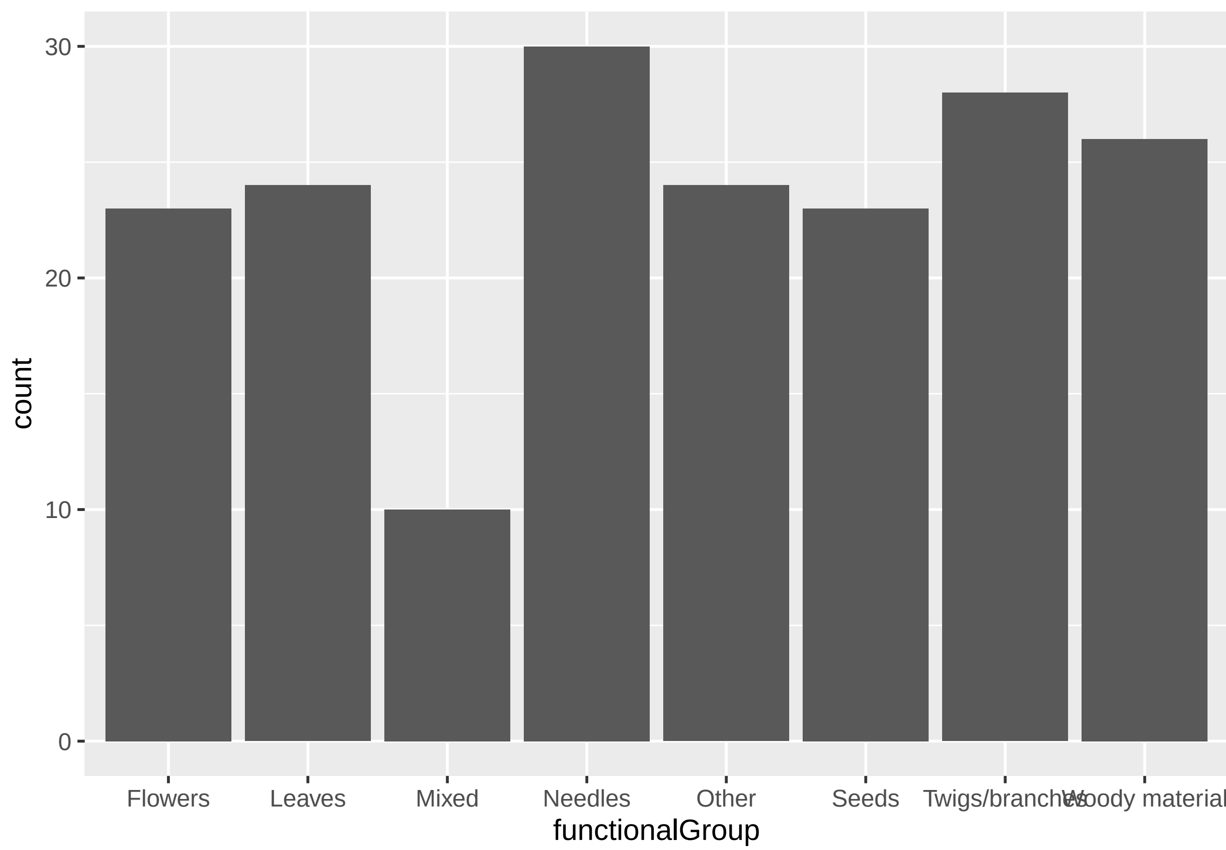
```r
#Listing unique plotIDs
unique(Litter$plotID)
```

```
##  [1] NIWO_061 NIWO_064 NIWO_067 NIWO_040 NIWO_041 NIWO_063 NIWO_047 NIWO_051
##  [9] NIWO_058 NIWO_046 NIWO_062 NIWO_057
## 12 Levels: NIWO_040 NIWO_041 NIWO_046 NIWO_047 NIWO_051 NIWO_057 ... NIWO_067
```

Answer: The 'unique' command only returns a list of different values found in the plotID column. The 'summary' command calculates and returns the number of entries that contain each unique plotID value.

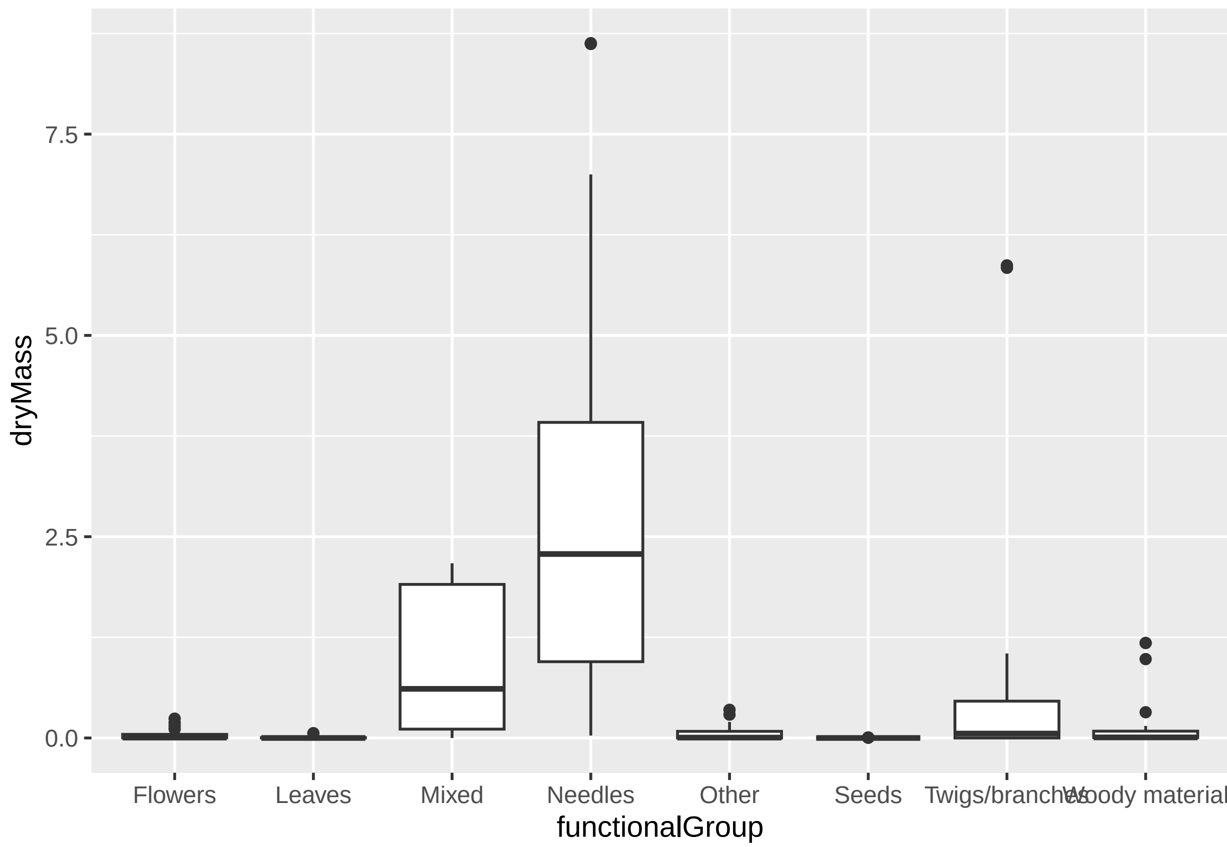
14. Create a bar graph of functionalGroup counts. This shows you what type of litter is collected at the Niwot Ridge sites. Notice that litter types are fairly equally distributed across the Niwot Ridge sites.

```r
#plotting functionalGroup as bar graph
ggplot(Litter) +
  geom_bar(aes(x=functionalGroup))
```

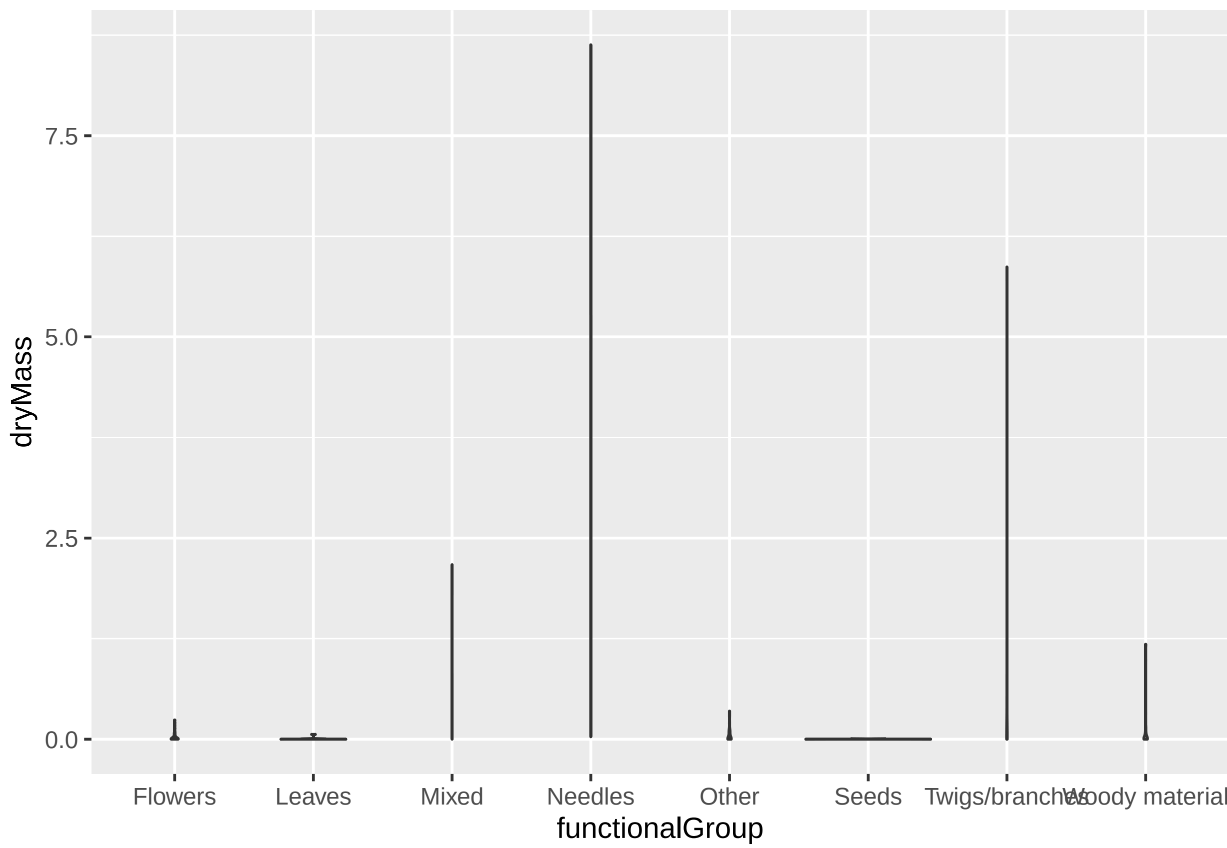


15. Using `geom_boxplot` and `geom_violin`, create a boxplot and a violin plot of dryMass by functionalGroup.

```
#plotting dryMass by functional group as boxpplot
ggplot(Litter) +
  geom_boxplot(aes(x=functionalGroup,y=dryMass))
```




```
#plotting dryMass by functional group as violin plot
ggplot(Litter) +
  geom_violin(aes(x=functionalGroup,y=dryMass))
```



Why is the boxplot a more effective visualization option than the violin plot in this case?

> Answer: The boxplot is more effective here because it is independent of density. The data in this set is too sparse to effectively visualize its density in a violin plot.

What type(s) of litter tend to have the highest biomass at these sites?

> Answer: Needles tend to have the highest biomass.